Derivatives Basics

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3. Bounds on Option Prices
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Definition (Derivatives)

Derivatives are financial securities whose payoffs are explicitly given by the value of some other underlying asset.

Examples

Important examples of derivatives are forward contracts, futures contracts, options, swaps, swaptions.

- The derivatives market is often estimated at more than $1.2 quadrillion (1200 trillion).
- Some reports says that the derivatives market is more than 10 times the size of the total world GDP.

Forward Contracts

Forward contract on an asset is a contract to buy or sell a unit amount of the asset at a specific price and at a fixed time in the future.

Example

Consider a Forward contract to buy 100,000 kg of sugar at 80 Rs per kg on 15th July.
Some features of Forward contracts:

- The contract is between two parties.
- The asset involved in the forward contract is called the **underlying asset**.
- **Buyer** is said to be in **long position** of the Forward contract.
- **Seller** is said to be in **short position**.
- The agreed price of the asset in the contract is called the **delivery price**.
- The time of trading of the asset in the contract is called **maturity**.
- The market for immediate delivery of the asset is called the **Spot Market**.
- The price of the asset in the spot market is called **Spot Price**.

Cash-flow from forward contract:

- Holding a long position in a forward contract with forward price F and spot price at T of the underlying asset S_T is given by $(0, S_T - F)$, here time points are $t = 0, T$.
- Similarly, cash flow of short position in the Forward contract is $(0, F - S_T)$.
- Note that zero value for entering into the forward contract is due to the fact that the payoff at maturity T of the forward contract can be negative also.

No Arbitrage Assumption

Definition (Arbitrage)

The simultaneous buying and selling of a security at two different prices in two different markets, resulting in profits without risk. Perfectly efficient markets present no arbitrage opportunities. Perfectly efficient markets seldom exist, but, arbitrage opportunities are often precluded because of transactions costs.

Example

As a hypothetical example, if Infosys stock trades at Rs 790 on the National Stock Exchange (NSE) and at Rs 800 on the Bombay Stock Exchange (BSE), an investor could guarantee a profit by purchasing the stock on the NSE and simultaneously selling the same amount of stock on the BSE.

Long Position vs a Short Position in Equity

Long Position: If an investor has long positions, it means that the investor has bought and owns those shares of stocks.

Short Position: If the investor has short positions, it means that the investor owes those stocks to someone, but does not actually own them yet. Sometime this is also called **short selling** which we mean "borrow the asset and sell it in the spot market to be returned later".

Example

Consider an investor who has sold 100 shares of Google without yet owning those shares is said to be short 100 shares. The short investor owes 100 shares at settlement and must fulfill the obligation by purchasing the shares in the market to deliver.

Some Assumptions

Our current assumptions of the model for the Market are the following:

- No arbitrage.
- There is an ideal bank with a spot rate curve $s_t, t = 1, 2, \cdots$.
- Among other securities, Market contains the underlying asset.
- Short selling of the underlying asset is allowed.

Spot Rates

- **Spot rate** for maturity t denoted by s_t is defined as the yield (per annum) on zero coupon bond with maturity t.
- Thus the term structure of interest rates corresponding to firms belonging to a category is given by Spot rate curve is given by $\{s_t, t = 1, 2, \cdots\}$ where t denote beginning of each period.
- For example, if periods are divided semi annually, then $t = 0, 1, \cdots$ corresponds to the time points today, 6 months, 12 months, 18 months etc.

Discount Rates and PV

- Discount rate d_t corresponding to the spot rate s_t is given by

$$d_t = \frac{1}{(1+s_t)^t} \quad \text{(annual compounding)}$$

- Note that Rs. 1 today becomes $(1+s_t)^t$ at the end of t years.
- Similarly,

$$d_t = \frac{1}{(1+s_t/2)^{2t}} \quad \text{(semi-annual compounding)}$$

- Moreover $d_t = e^{-s_t t}$ (continuous compounding)
- The present value (with respect to the spot rate curve $\{s_t, t = 0, 1, \cdots\}$) of the cash flow $(x_0, x_1, \cdots, x_n)$ is

$$PV = x_0 + d_1 x_1 + \cdots + x_n d_n.$$

Example: constructing the spot curve

Market Data :	6 months bill		Price 97.44
	1 year bill		Price 94.88
	18 months Bond	coupon 5%	Price 99.38
	2 year Bond	coupon 6%	Price 100.89

Semi-annual coupon payment and all have face value \$ 100 .

$$97.44 = \frac{100}{(1+s_{.5}/2)}, \quad s_{.5} = 5.25\%.$$

$$94.88 = \frac{100}{(1+s_1/2)^2}, \quad s_1 = 5.32\%.$$

$$99.38 = \frac{2.5}{(1+s_{.5}/2)} + \frac{2.5}{(1+s_1/2)^2} + \frac{102.5}{(1+s_{1.5}/2)^3}, \quad s_{1.5} = 5.44\%.$$

$$100.89 = \frac{3}{(1+s_{.5}/2)} + \frac{3}{(1+s_1/2)^2} + \frac{3}{(1+s_{1.5}/2)^3} + \frac{103}{(1+s_2/2)^4}, \quad s_2 = 5.53\%.$$

Relation between current price and forward price

Theorem

Let F denote the forward price, T maturity time, $d(0,T)$ the discount factor for the period $[0,T]$ and S_0 current spot price of the underlying asset. Also assume that there is no storage cost or dividend for the underlying asset. Then

$$F = \frac{S_0}{d(0,T)}.$$

Proof.

The proof is based on the "no arbitrage argument". Suppose

$$F > \frac{S_0}{d(0,T)}$$

- Borrow S_0 from Bank. [Cash flow $(S_0, -\frac{S_0}{d(0,T)})$].
- Buy unit asset from the spot market and store. [Cash flow $(-S_0, S_T)$]
- Short unit forward contract. [Cash flow $(0, F - S_T)$].

This creates an "arbitrage". This can be seen from the following observation. Initial value of the above investment is zero and it has a final payoff of $F - \frac{S_0}{d(0,T)}$, an arbitrage opportunity. $\square$

Forward rate in FX market: Covered Interest Rate Parity

The forward foreign exchange rate are given by

$$F = S\frac{(1 + i_f)}{(1 + i_d)},$$

where

- i_d =The interest rate in the domestic currency or the base currency
- i_f = The interest rate in the foreign currency or the quoted currency
- S = The current spot exchange rate
- F = The forward foreign exchange rate

Forward contract with storage cost

- Now we assume the underlying asset has carrying cost c_k for the period $[k, k+1), k = 0, 1, \cdots, M$ payable at the beginning of the period.
- M is the number of periods until maturity of the forward contract.
- If underlying is a commodity, then carrying cost (storage cost) can be cost to store or insurance fee etc. which in general positive.
- If the underlying is a financial security such as stock, carrying is cost is dividend and hence is negative.

Theorem

The forward price assuming carrying cost is given by

$$F = \frac{S_0}{d(0,M)} + \sum_{k=0}^{M-1} \frac{c_k}{d(k,M)}.$$

Proof.

Let

$$F < \frac{S_0}{d(0,M)} + \sum_{k=0}^{M-1} \frac{c_k}{d(k,M)}.$$

Then

- Short sell one unit of underlying asset and receive S_0 at time 0.
- Invest S_0 in Bank at time 0.
- Long one unit Forward contract at time 0.
- Receive c_k at the beginning of the period $[k, k+1)$ and invest in Bank.

Clearly net value of the above investment at 0 is 0. The value at time T is

$$\frac{S_0}{d(0,M)} + \frac{c_0}{d(0,M)} + \frac{c_1}{d(1,M)} + \cdots + \frac{c_{M-1}}{d(M-1,M)} - F > 0.$$

Therefore, the above investment strategy is an arbitrage. $\square$

Value of a Forward contract

- Forward price varies with time. i.e. it is a function of time.
- Let F_t denote the forward price at time t.
- We have the following question. Suppose you are holding a long forward contract at time 0 and current time is t. What is the value of your contract today? i.e. suppose you would like to transfer your contract to a second party, how much she will pay(+ or -) for it.
- The answer is: value of the contract, $f_t = (F_0 - F_t)d(t, T)$.

Proof

- Consider the portfolio of one long forward contract with delivery price F_t and a short forward contract with delivery price F_0 at time t.
- The cash flow for the long position is

$$(0, S_T - F_t).$$

- The cash flow from the short position is $(-f_t, F_0 - S_T)$
- The net cash flow from the portfolio is $(-f_t, F_0 - F_t)$
- The portfolio has a deterministic cash flow and hence

$$f_t = (F_0 - F_t)d(t, T).$$

Example

A 10 year 16% bond with face value Rs. 100 is currently selling for Rs. 92. A long forward contract on this bond that has a delivery date in 1 year and a delivery price of Rs. 94. The bond pays coupons semi annually with one due 6 months from now and another just before the maturity of the forward contract. The current interest rates (compounded semi annually) for 6 months and 1 year are 7% and 8% respectively. Find the current value of the forward contract.

Solution. We have

$$d(0,1/2) = \frac{1}{1.035} = 0.966; d(0,1) = \frac{1}{1.04^2} = 0.924; d(1/2;1) = \frac{d(0,1)}{d(0,1/2)} = 0.956.$$

Delivery price $K = 94$. Let F_0 denote the current forward rate. Then price of the long forward contract with delivery price K is $f_0 = (F_0 - K)d(0,1)$. Let us calculate F_0.

$$F_0 = \frac{92}{d(0,1)} + \frac{-8}{d(1/2,1)} - 8 = 99.57 - 8.37 - 8 = 83.2.$$

Hence $f_0 = (83.2 - 94) \times 0.924 = -9.98$.

Futures Contract

- Futures contract is like forward contract with the difference of being traded in an organized exchange.
- Futures contract is designed to eliminate the default risk associated with forward contract.
- It is achieved through the process called marking to the market.

Futures Contract Contd...

- While entering into the futures contract both parties need to deposit an amount with the broker called the margin account.
- At the end of each day, the net receipt will be marked. i.e. at the end of each day, the delivery price will be revised to the current delivery (futures price) and the net receipt from this process is added to the market.
- For example, at the end of day one, for the party holding the long position in the futures contract $F_1 - F_0$ will be deposited in her/his margin account, where F_0 is the futures price at the time of entry of the contract and F_1 is the futures price at the end of day one.
- when the margin value falls below a certain level, the contract holder need to pay money to makeup the margin account.

Futures Contract on S&P 500

Underlying	S&P 500 index
Where traded	Chicago Mercantile Exchange
Size	$250 × S&P 500 index
Months	March, June, September, December
Trading ends	Business day prior to determination of settlement price
Settlement	Cash-settled, based upon opening price of S&P 500 on third Friday of expiration month

Margins and Marking to Market

- Suppose the futures price is 1100 and you wish to acquire a \$2.2 million position in the S&P 500 index.
- The notional value of one contract is $\$250 \times 1100 = \$275,000$; this represents the amount you are agreeing to pay at expiration per futures contract.
- To go long \$2.2 million of the index, you would enter into \$2.2 million/\$0.275 million = 8 long futures contracts.
- Both buyers and sellers are required to post a security deposit with the broker to ensure that they can cover a specified loss on the position. This deposit, which can earn interest, is called **margin** and is intended to protect the counterparty against your failure to meet your obligations.

MoM Contd...

- suppose that there is 10% margin and daily settlement. The margin on futures contracts with a notional value of \$2.2 million is \$220,000.
- If the S&P 500 futures price drops by 1, to 1099, we lose \$2000 on our futures position.
- Thus, if the continuously compounded interest rate is 6%, our margin balance after 1 day is

$$220,000e^{0.06/365} - 2000 = 220036.17 - 2000 = 218036.17.$$

- The decline in the margin balance means the broker has significantly less protection should we default. For this reason, participants are required to maintain the margin at a minimum level, called the maintenance margin. This is often set at 70% to 80% of the initial margin level.
- The broker make a **Margin Call**.

Definition (Option)

An option is a financial contract where the buyer of the contract has the right to buy/sell specific number of units of the underlying asset at a specified price on or before a Specified date. This right is granted by the seller of the option.

Terminologies related to options

- Buyer of the option is called holder and seller is called writer.
- The specific time till the option is valid is called the maturity.
- Specified price = **strike** or **exercise price**
- Buyer = holder = long position
- Seller = writer = short position
- If the holder has the write to buy, it is called **call option**.
- If the holder has the write to sell, it is called **put option**.
- On/before: **American**. Only on expiration: **European**.

In the Money and Out of the Money Options

- A call option with strike K is said the be **in the money option** if $S_0 > K$.
- Call option is called **out of the money** if $S_0 < K$.
- It is called **at the money** if $S_0 = K$.
- Here S_0 is the current price of the underlying asset.
- The payoff at maturity of a European call option is

$$(S_T - K)^+ = \max\{S_T - K, 0\},$$

where S_T is the price of the underlying at the maturity T.

TCS Option Chain

TCS | OPTIONS CHAIN | NIFTY

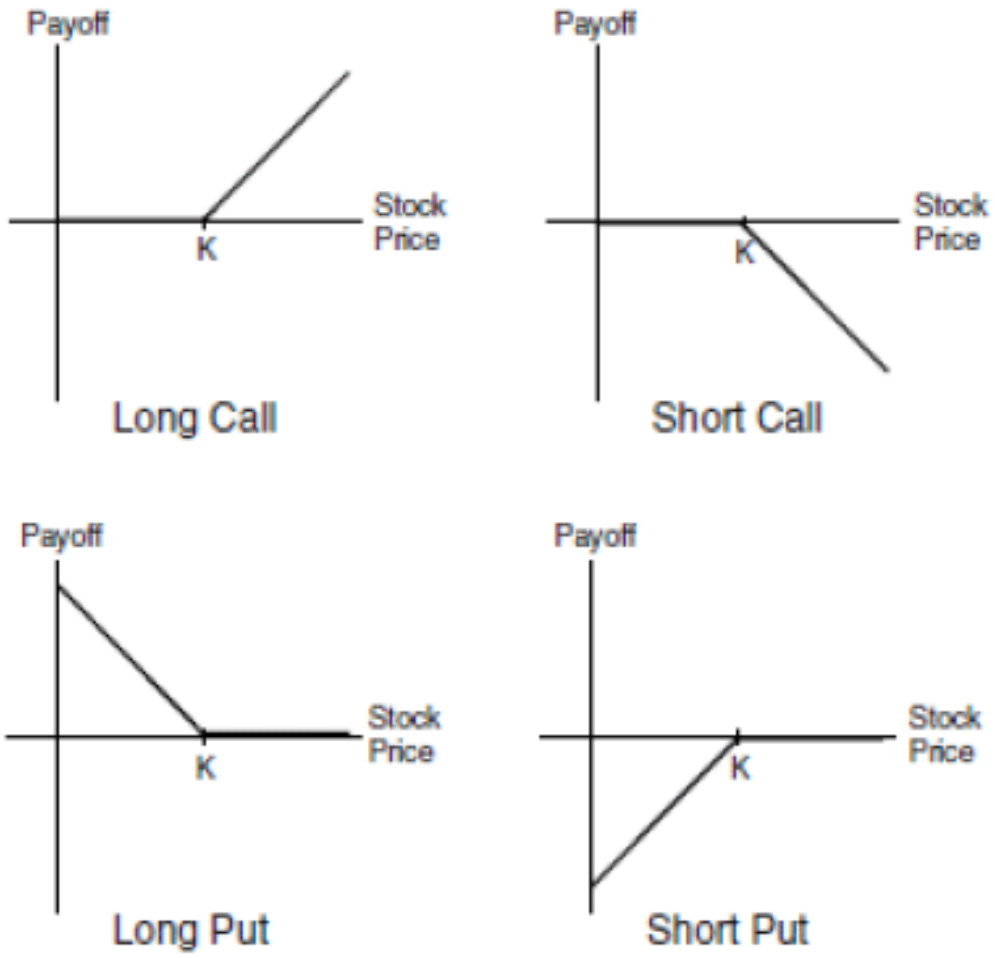
Expiry Dates (TCS) : 28APR2022

Expiry Dates (NIFTY) : --Select--

TCS Expiry Date 28APR2022 — **As on 02-FEB-2022 16:00:00 Hours IST**

Calls											Puts									
Quote	Open Interest	Change in Open Interest	LTP	Net Change	Volume	Bid Qty	Bid Price	Offer Price	Offer Qty	Strike Price	Bid Qty	Bid Price	Offer Price	Offer Qty	Volume	Net Change	LTP	Open Interest	Change in Open Interest	Quote
Quote	-	-	-	-	-	-	-	-	-	3300.00	-	-	30.00	150	-	-	-	-	-	Quote
Quote	-	-	-	-	-	-	-	-	-	3400.00	-	-	30.00	150	-	-	-	-	-	Quote
Quote	-	-	-	-	-	-	-	-	-	3500.00	150	31.00	50.95	150	9	-8.00	37.00	1,050	900	Quote
Quote	-	-	-	-	-	-	-	-	-	3600.00	300	45.00	57.90	150	4	-111.20	55.15	300	300	Quote
Quote	-	-	-	-	-	-	-	-	-	3700.00	150	75.00	-	-	-	-	-	-	-	Quote
Quote	-	-	-	-	-	150	103.00	-	-	3800.00	150	121.05	129.65	150	12	-12.00	123.00	1,800	1,650	Quote
Quote	-	-	-	-	-	150	73.80	-	-	3900.00	150	127.00	-	-	-	-	-	-	-	Quote
Quote	9,750	2,550	135.00	12.45	23	150	132.45	135.55	150	4000.00	150	171.00	325.00	150	-	-	-	-	-	Quote
Quote	-	-	-	-	-	150	45.00	-	-	4060.00	-	-	-	-	-	-	-	-	-	Quote
Quote	150	-	-	-	-	150	89.00	100.55	150	4100.00	-	-	-	-	-	-	-	-	-	Quote
Quote	-	-	-	-	-	150	41.00	80.55	150	4200.00	-	-	-	-	-	-	-	-	-	Quote
Quote	-	-	-	-	-	150	30.00	100.00	150	4300.00	-	-	-	-	-	-	-	-	-	Quote
Quote	150	150	34.00	3.50	10	1,200	28.00	39.00	150	4400.00	-	-	-	-	-	-	-	-	-	Quote

Call Option Payoff



Factors which Affect Option Price

- **Current price of the underlying:** bigger the current price bigger the call option price.
- **Strike price:** bigger the strike price smaller the call option price.
- **Volatility:** high volatility implies high option price for both call and put.
- **Maturity:** large time to maturity imply high option price.

Example

Today's price of a particular stock is Rs 120 and you expect that stock price will go significantly up in next 3 months. Suppose the 3-months call option with strike Rs 121 has price Rs 3. In this situation an investor has two choices either to buy the underlying equity or take a long position in the call option. Let us further assume that stock price become 125 at the end of 3 months. The stock return in 3 months is

$$\frac{125 - 120}{120} = 4.16\%.$$

The return from call option (in 3 months) will be

$$\frac{4 - 3}{3} = 33.33\%.$$

Thus investing in call option gives a **leveraged position**.

Covered Call

Covered Call: Portfolio of a long position in the underlying asset and a short position in the call option.

- The main goal of the covered call is to collect income via option premiums by selling calls against a stock that you already own.
- Assuming the stock doesn't move above the strike price, you collect the premium and maintain your stock position.
- Covered call writing is typically used by investors and longer-term traders, and is rarely used by day traders.

Example

Suppose you own 100 shares of XYZ corp. XYZ corp 3-months call option price with strike 110 is Rs 5. A covered call strategy in this scenario would be to have a short position in the call option. Covered call are generally employed in OTM options.

Bull Spread

Portfolio of a long position in call option with strike K_1 and a short position in call option with strike K_2, where $K_1 < K_2$. Payoff at maturity T is given by

$$\begin{cases} 0 & \text{if } S_T \leq K_1 \\ S_T - K_1 & \text{if } K_1 < S_T \leq K_2 \\ K_2 - K_1 & \text{if } S_T > K_2. \end{cases}$$

If **both the calls are OTM**, then

- Cost of buying the bull spread is low.
- Small chance of receiving the maximum payoff $K_2 - K_1$.

If **both call are ITM**, then

- Cost of buying the bull spread is high.
- High probability of receiving the maximum payoff $K_2 - K_1$.

Bear Spread

Portfolio of a short position in put option with strike K_1 and a long position in put option with strike K_2, where $K_1 < K_2$. Payoff at maturity T is given by

$$\begin{cases} K_2 - K_1 & \text{if } S_T \leq K_1 \\ K_2 - S_T & \text{if } K_1 < S_T \leq K_2 \\ 0 & \text{if } S_T > K_2. \end{cases}$$

If **both the puts are OTM**, then

- Cost of buying the bear spread is low.
- Small chance of receiving the maximum payoff $K_2 - K_1$.

If **both puts are ITM**, then

- Cost of buying the bear spread is high.
- High probability of receiving the maximum payoff $K_2 - K_1$.

Butterfly Spread

Portfolio of long positions in two call options with strikes K_1, K_3 and two short positions in a call option with strike K_2, where $2K_2 = K_1 + K_3$. Payoff at maturity T is given by

$$\begin{cases} 0 & \text{if } S_T \leq K_1 \\ S_T - K_1 & \text{if } K_1 < S_T \leq K_2 \\ 2K_2 - K_1 - S_T & \text{if } K_2 < S_T \leq K_3 \\ 2K_2 - K_1 - K_3 = 0 & \text{if } S_T > K_3. \end{cases}$$

Bounds on call options

Assume that underlying asset doesn't pay dividend and c_0 and S_0 are the prices of the European call and underlying asset at time 0 respectively. The following bounds hold.

$$S_0 - Ke^{-rT} \leq c_0 \leq S_0.$$

We prove it by contradiction.

1. Suppose $c_0 > S_0$. Consider the following portfolio.
   - Write a call and receive c_0.
   - Buy the asset at the spot price S_0.
   - Invest $(c_0 - S_0)$ in the bank at the rate r.
2. The cashflow from this portfolio at time 0 is 0.
3. The cashflow at time T is

$$\begin{cases} S_T + (c_0 - S_0)e^{rT} & \text{if } S_T \leq K \text{ (call not exercised)} \\ K + (c_0 - S_0)e^{rT} & \text{if } S_T > K \text{ (call exercised).} \end{cases}$$

4. Thus portfolio value is positive at time T, with 0 investment in beginning, which is an arbitrage opportunity. Thus $c_0 > S_0$ is not possible, hence $c_0 \leq S_0$.

Bounds on call option contd...

Assume that $c_0 < S_0 - Ke^{-rT}$. Then consider the following portfolio at 0.

- Short sell the asset and generate S_0.
- Buy the call after paying c_0.
- Invest $(S_0 - c_0)$ in the bank at the rate r.

1. The cashflow from this portfolio at time 0 is 0.
2. The cashflow at time T is

$$\begin{cases} -S_T + (S_0 - c_0)e^{rT} & \text{if } S_T \leq K \text{ (call not exercised)} \\ -K + (S_0 - c_0)e^{rT} & \text{if } S_T > K \text{ (call exercised).} \end{cases}$$

3. Thus portfolio value is positive at time T, with 0 investment in beginning, which is an arbitrage opportunity. Thus $c_0 \geq S_0 - Ke^{-rT}$.

Put-call parity

Let c_t and p_t denote respectively price at time t of a European call and European put with strike K respectively. Also let S_t be the price of the asset at time t. The following holds true

$$c_t + Ke^{-r(T-t)} = p_t + S_t, 0 \leq t \leq T.$$

- We will prove the parity for $t = 0$, other cases also follows in similar way.
- Suppose $c_0 + Ke^{-rT} > p_0 + S_0$.
- Consider the following portfolio of assets
  - Write (sell) a call option and borrow Ke^{-rT}
  - Buy a put option and also buy a unit of the underlying asset.
  - Invest $c_0 + Ke^{-rT} - p_0 - S_0$ in the bank.
- The cashflow from this portfolio at time 0 is 0.

Put-call parity contd...

- The cashflow from this portfolio at time 0 is 0.
- The cashflow at time T is

$$\begin{cases} -K + K + (c_0 + Ke^{-rT} - p_0 - S_0)e^{rT} & \text{if } S_T \leq K \text{ (put exercised)} \\ K - K + (c_0 + Ke^{-rT} - p_0 - S_0)e^{rT} & \text{if } S_T > K \text{ (call exercised).} \end{cases}$$

- Thus portfolio value is positive at time T, with 0 investment in beginning, which is an arbitrage opportunity. Thus $c_0 + Ke^{-rT} \leq p_0 + S_0$.
- Similarly we prove that $c_0 + Ke^{-rT} \geq p_0 + S_0$.

Exercise: Verify the Put-Call Parity and Call option Bound for $K = 1600$

Expiry Dates (INFY) : 24FEB2022

Expiry Dates (NIFTY) : --Select--

INFY **Expiry Date 24FEB2022**

As on 07-FEB-2022 16:00:00 Hours IST

Calls											Puts									
Quote	Open Interest	Change in Open Interest	LTP	Net Change	Volume	Bid Qty	Bid Price	Offer Price	Offer Qty	Strike Price	Bid Qty	Bid Price	Offer Price	Offer Qty	Volume	Net Change	LTP	Open Interest	Change in Open Interest	Quote
Quote	-	-	-	-	-	12,000	520.65	670.15	12,000	1120.00	300	0.10	1.75	12,000	-	-	-	-	-	Quote
Quote	-	-	-	-	-	12,000	503.65	635.20	12,000	1140.00	6,000	0.05	0.90	300	-	-	-	300	-	Quote
Quote	-	-	-	-	-	12,000	486.25	625.90	12,000	1160.00	6,000	0.05	3.00	12,000	-	-	-	-	-	Quote
Quote	-	-	-	-	-	12,000	467.05	591.30	12,000	1180.00	6,000	0.05	3.00	12,000	-	-	-	-	-	Quote
Quote	15,900	-	-	-	-	12,000	459.55	566.45	12,000	1200.00	2,700	0.30	0.50	8,100	1	-	0.30	17,100	-	Quote
Quote	-	-	-	-	-	12,000	433.35	545.95	12,000	1220.00	6,000	0.10	3.00	12,000	-	-	-	-	-	Quote
Quote	-	-	-	-	-	12,000	417.05	523.15	12,000	1240.00	6,000	0.10	2.35	12,000	-	-	-	-	-	Quote
Quote	-	-	-	-	-	12,000	398.80	504.10	12,000	1260.00	6,000	0.10	3.00	12,000	-	-	-	-	-	Quote
Quote	-	-	-	-	-	12,000	390.15	481.50	12,000	1280.00	6,000	0.10	3.05	1,200	-	-	-	-	-	Quote
Quote	-	-	-	-	-	12,000	364.60	459.60	12,000	1300.00	300	0.30	1.25	3,000	-	-	-	1,800	-	Quote
Quote	-	-	-	-	-	12,000	355.45	439.75	12,000	1320.00	6,000	0.10	3.00	600	-	-	-	-	-	Quote
Quote	-	-	-	-	-	12,000	335.00	408.50	12,000	1340.00	6,000	0.10	3.00	600	-	-	-	-	-	Quote
Quote	-	-	-	-	-	12,000	319.15	387.35	12,000	1360.00	6,000	0.10	3.00	600	-	-	-	-	-	Quote
Quote	-	-	-	-	-	12,000	300.60	392.35	12,000	1380.00	6,000	0.10	3.00	600	-	-	-	-	-	Quote
Quote	2,400	-	-	-	-	900	288.95	351.35	12,000	1400.00	4,500	0.90	1.00	1,800	145	-	0.90	99,600	300	Quote
Quote	-	-	-	-	-	900	262.95	339.70	12,000	1420.00	6,000	0.10	-	-	-	-	-	300	-	Quote
Quote	-	-	-	-	-	900	243.05	311.80	12,000	1440.00	6,000	0.10	-	-	-	-	-	-	-	Quote
Quote	-	-	-	-	-	900	223.35	292.20	12,000	1460.00	300	0.25	-	-	-	-	-	-	-	Quote
Quote	-	-	-	-	-	900	203.25	270.00	12,000	1480.00	300	1.05	3.00	900	3	-17.10	2.90	-	-	Quote
Quote	9,300	900	215.00	-13.70	6	900	203.35	214.60	600	1500.00	300	2.90	3.20	10,200	779	1.10	3.20	382,200	29,100	Quote
Quote	-	-	-	-	-	1,200	163.75	216.30	900	1520.00	300	1.80	-	-	-	-	-	15,300	-	Quote
Quote	-	-	-	-	-	12,900	138.75	210.25	12,000	1540.00	300	4.20	4.55	300	568	1.60	4.40	187,800	39,000	Quote
Quote	-	-	-	-	-	1,200	136.25	191.35	12,000	1560.00	600	3.00	5.85	300	927	2.10	5.55	123,900	31,500	Quote
Quote	-	-	-	-	-	900	126.45	146.30	900	1580.00	300	4.30	8.95	300	610	2.55	6.75	42,300	-600	Quote
Quote	38,700	-14,700	118.00	-32.10	226	600	116.45	119.50	300	1600.00	300	8.45	8.50	3,900	2,212	3.10	8.50	465,000	56,400	Quote
Quote	1,500	300	103.45	-26.10	2	300	92.20	101.80	600	1620.00	300	11.00	11.30	300	1,701	4.30	11.15	153,600	52,800	Quote
Quote	20,700	-600	83.00	-30.10	23	900	71.35	85.80	600	1640.00	300	14.50	14.90	300	1,972	5.75	14.75	314,700	127,500	Quote
Quote	43,800	-2,100	68.00	-26.95	188	300	66.30	74.25	300	1660.00	300	18.55	19.80	300	2,280	7.80	19.75	306,000	17,700	Quote
Quote	49,500	10,800	55.00	-24.55	782	600	54.50	57.90	300	1680.00	300	21.20	27.40	300	2,151	9.25	25.00	179,700	5,700	Quote

Example

The price of a non-dividend paying stock is Rs 190 and the price of a 3-month European call option on the stock with strike Rs 200 is Rs 10. The risk-free rate is 4% per annum compounded continuously. What is the price of a 3-month European put option on the stock with strike Rs 200.

Solution. We have $c_0 = 10$; $K = 200$; $S_0 = 190$; $r = 0.04$; $T = 0.25$. Using put-call parity, we get $p_0 = 18.01$.

Verifying put call parity with NSE option chains data

Example

Consider Infosys NSE option chains data with expiry March 26, 2020, with strike price 700. We take mid prices i.e. the average of the bid and ask prices to remove the effect of bid-ask spread. We have

- $c_t = \frac{78.25+92.80}{2} = 85.525$
- $p_t = \frac{1.55+9.45}{2} = 5.5$
- $K = 700$
- $T - t = 44/365 = 0.12055$
- $r = 0.0514$
- $S_t = 776$
- Thus $c_t + Ke^{-r(T-t)} = 780.41$
- $p_t + S_t = 781.5$
- Thus put-call parity holds approximately.

Some References

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THANK YOU